

SHARP WASSERSTEIN PROPAGATION OF CHAOS FROM CORRELATED INITIAL DATA

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ABSTRACT. We study mean field diffusions with a globally Lipschitz one-body drift and a bounded, globally Lipschitz pairwise interaction. Suppose that every k -particle marginal of the initial law lies within squared quadratic Wasserstein distance $C_0(k/N)^2$ of the corresponding power of the limiting law. We prove that the same holds on every finite time interval, with an explicit constant that depends neither on N nor on the dimension. The initial law may be correlated and singular, as for the uniform distribution on Kac's sphere, and neither an entropy bound nor a transport inequality is assumed. For bounded interactions this answers the time-marginal version of a question of Lacker. The proof combines an entropy-cost regularization of an independent reference dynamics with a BBGKY-type hierarchy for Wasserstein tangent energies, which is closed by comparison with a Yule process.

1. INTRODUCTION

Consider N particles in $\mathbb{R}^d$ evolving according to

$$(1.1) \quad dX_i^N(t) = \left(a(X_i^N(t)) + \frac{1}{N} \sum_{j=1}^N K(X_i^N(t), X_j^N(t)) \right) dt + \sqrt{2} dW_i(t), \quad 1 \leq i \leq N,$$

where $W_1, \dots, W_N$ are independent Brownian motions and the initial law $P_{N,0}$ on $(\mathbb{R}^d)^N$ is exchangeable. As $N \rightarrow \infty$, each particle is expected to follow the McKean–Vlasov equation

$$(1.2) \quad dX(t) = B_t(X(t))dt + \sqrt{2} dW(t), \quad \mu_t = \text{Law}(X(t)),$$

where $B_t(x) = a(x) + \int K(x, y) \mu_t(dy)$. Let $P_{N,t}$ be the law of (1.1) at time t and $P_{N,t}^{(k)}$ the joint law of the first k particles. Propagation of chaos means that $P_{N,t}^{(k)}$ is close to $\mu_t^{\otimes k}$ whenever this holds at time zero [34, 44, 24, 58]. This paper is concerned with the size of $W_2(P_{N,t}^{(k)}, \mu_t^{\otimes k})$, as a function of k and N , when the initial law is correlated and possibly singular.

For Lipschitz coefficients and independent initial positions, synchronous coupling, which drives the particles and N independent copies of (1.2) by the same Brownian motions, gives $W_2^2(P_{N,t}^{(k)}, \mu_t^{\otimes k}) = O(k/N)$ on bounded time intervals [58]. The optimal order is $(k/N)^2$, as the Gaussian example of [39, Example 2.8] shows. Two particles are correlated at order $1/N$, so the covariance of k particles differs from that of $\mu_t^{\otimes k}$ by a perturbation of size k/N in the direction of $(1, \dots, 1)$, and W_2^2 is quadratic in such a perturbation; Proposition 2.7 computes this exactly for a Gaussian sample recentred at its empirical mean. Lacker [39] proved that the relative entropy of the law of the first k trajectories on $[0, T]$ with respect to the law of k independent solutions of (1.2) is $O((k/N)^2)$ whenever the same bound holds at time zero; in particular $H(P_{N,t}^{(k)} | \mu_t^{\otimes k}) = O((k/N)^2)$. He obtained this by comparing consecutive marginals

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through the BBGKY hierarchy, and deduced the corresponding path-space bound for W_2^2 for Lipschitz coefficients from a quadratic transport inequality for the limit [59, 50, 26]. After [39, Corollary 2.7] he asked whether this path-space Wasserstein bound can be propagated from its analogue at time zero, without assuming entropic chaos initially.

We answer this question for time marginals when the interaction is bounded. Theorem 2.1 states that if a and K are Lipschitz, K is bounded, and

$$(1.3) \quad W_2^2(P_{N,0}^{(k)}, \mu_0^{\otimes k}) \leq C_0 \frac{k^2}{N^2}, \quad 1 \leq k \leq N,$$

then $W_2^2(P_{N,t}^{(k)}, \mu_t^{\otimes k}) \leq C_T k^2 / N^2$ for $0 \leq t \leq T$ and $1 \leq k \leq N$. The constant C_T is explicit and depends only on C_0 , T , the Lipschitz constants of a and K , and $\sup |K|$. It does not depend on N , on the dimension, or on moments of the initial laws.

Condition (1.3) involves no entropy. It holds for the microcanonical laws of Section 2.4: the uniform distribution on the sphere $\{\sum_i |x_i|^2 = dN\}$, which for $d = 1$ is Kac's sphere [34], and the law of a Gaussian sample recentred at its empirical mean. Both laws are singular with respect to $\mu_0^{\otimes N}$, so the entropic hypothesis of [39] fails at level N . For independent initial positions, Theorem 2.1 only requires a finite second moment of μ_0 , whereas the passage from entropy to W_2 in [39, Corollary 2.7] uses a quadratic transport inequality for μ_0 . The order $(k/N)^2$ also controls correlations: for bounded Lipschitz f , the covariance of $f(X_1^N(t))$ and $f(X_2^N(t))$ is $O(1/N)$ (Corollary 2.3), while the same argument based on the order k/N gives $O(N^{-1/2})$. Interactions that grow in the second variable, such as $K(x, y) = y - x$, are not covered; we return to them in Section 8.

1.1. Main ideas. Lacker's argument rests on the chain rule for relative entropy: the increment $H(P^{(k+1)} | \mu^{\otimes(k+1)}) - H(P^{(k)} | \mu^{\otimes k})$ is the averaged entropy of the conditional law of a further particle given k others, and this conditional law is what enters the drift of the k -particle marginal. Wasserstein distances have no such structure. If X_1 is uniform on $[0, 1]$ and X_2 is the fractional part of MX_1 , then as $M \rightarrow \infty$ the law of (X_1, X_2) converges weakly, hence in W_2 on the unit square, to the uniform law, while the conditional law of X_2 given X_1 is a Dirac mass for every M . Closeness of joint laws in W_2 gives no control of conditional laws.

We therefore keep the Wasserstein distance for the comparison of the particle system with its limit, and use relative entropy only for an explicit reference law. Let R_s^N be the law at time s of N independent copies of the linear diffusion with drift B , started from $P_{N,0}$. It keeps the correlations of the initial law, while the noise regularizes each particle, and an entropy-cost inequality turns (1.3) into

$$H(R_s^{(k)} | \mu_s^{\otimes k}) \leq \frac{C_0 D_T}{s} \frac{k^2}{N^2}, \quad 0 < s \leq T,$$

where D_T depends only on T and the Lipschitz constants. The particle law is compared with R_t^N along the curve $\nu_s = S_{t-s}^{N,*} R_s^N$, $0 \leq s \leq t$, which follows the reference dynamics up to time s and the particle dynamics afterwards; here $S^{N,*}$ is the particle semigroup acting on measures. The velocity of this curve at time s is the difference of the two generators applied to R_s^N , transported by the linearized particle flow over the remaining time $t - s$. We measure the velocity of the k -particle marginal in the tangent space of the Wasserstein space at that marginal, that is, in a negative Sobolev norm weighted by the marginal itself, so that no density is needed. Since lengths are integrals of speeds, only the square root of the factor s^{-1} is integrated, and it is integrable at $s = 0$.

The order k^2/N^2 comes from conditioning. The generator difference is the vector field $U_s(x)_i = -N^{-1} \sum_j c_s(x_i, x_j)$, where c_s is the interaction centred with respect to μ_s . Under

$\mu_s^{\otimes N}$ each component of U_s has size of order $N^{-1/2}$, so that $\mathbb{E}|(U_s)_{1:k}|^2$ is of order k/N . Only the conditional expectation of $(U_s)_{1:k}$ given the first k particles acts on the k -particle marginal, and its mean square is $O(k^2/N^2)$ (Lemmas 4.5 and 4.6). The tangent energy of the k -particle marginal is the squared norm of the gradient part of this conditional expectation (Lemmas 3.2 and 3.3).

It remains to follow the tangent energies $E_k(r)$ of the velocity as it is transported. They satisfy a hierarchy of BBGKY type, which in differential form reads

$$E'_k \leq D E_k + q_k (E_{k+1} - E_k), \quad q_k = \lambda k \left(1 - \frac{k}{N}\right),$$

with constants D and λ depending only on the Lipschitz constants and $\sup |K|$, and which is proved in integrated form in Proposition 5.4. The coupling term $E_{k+1} - E_k$ is the squared norm of the part of the $(k+1)$ -particle representative that is orthogonal to the k -particle one (Lemma 3.3), so it is nonnegative. The inequality is the backward equation of a pure-birth process with rates q_k , which is dominated by a Yule process, and the profile k^2/N^2 is preserved up to a factor $e^{\omega r}$. Since the laws carrying the velocities may be singular, the hierarchy is derived from a Galerkin approximation by trigonometric polynomials (Section 5 and Appendix A).

1.2. Related work. We refer to the surveys of Chaintron and Diez [11, 12] for the many approaches to propagation of chaos and mention only work close to the present one. Coupling methods go back to Dobrushin [19] and Sznitman [58], and give time-uniform bounds under convexity, see for instance [43]. Relative entropy methods were developed by Jabin and Wang [32, 33] for bounded and $W^{-1,\infty}$ interactions; see also [37, 31, 23]. These works estimate the entropy of the whole configuration, and subadditivity then yields the order k/N for k -particle marginals.

The local approach of [39] has been extended to time-uniform estimates [40, 46], to L^p and singular interactions [28, 62], to higher-order expansions [30], to non-constant diffusion coefficients [27] and to nonlinear dependence on the empirical measure [3]. For non-exchangeable systems, Lacker, Yeung and Zhou [41] use a hierarchy indexed by subsets of particles together with first-passage percolation, and in the mean field case their comparison process is a Yule process [41, Section 1.5]. Sharp marginal bounds also enter the fluctuation theory of [57]. The size of correlations between particles is studied in [51, 22]. High-temperature mean field Gibbs measures satisfy the entropic analogue of (1.3) [38], and the creation of chaos from correlated data is studied in [54, 5]. Kac's chaos for measures on spheres is quantified in [29, 10]; see also [17] for the comparison of spherical and Gaussian marginals.

Differentiation along the curve ν_s gives a Duhamel formula, as in the consistency–stability method of Mischler, Mouhot and Wennberg [45] and in weak error expansions [13, 16]. Here the consistency error is evaluated on the explicit reference law R_s^N , and stability comes from the linearized particle flow, measured in marginal tangent energies. For the tangent space of the Wasserstein space we follow [49, 1]; comparisons between W_2 and weighted $\dot{H}^{-1}$ norms appear in [42, 52]. For Coulomb and Riesz interactions, modulated energy and modulated free energy methods [21, 56, 47, 8, 55, 14] exploit the structure of the interaction instead of Lipschitz bounds.

1.3. Organization. Section 2 contains the main theorem, its consequences and the examples. Section 3 introduces tangent energies, the reference evolution and the interpolating curve, and proves Theorem 2.1 from three intermediate results, which are established in Sections 4, 5 and 6; Section 7 contains the approximation lemmas with which Section 3.4 removes the

smoothness assumptions. Section 8 discusses extensions, and Appendix A contains the Galerkin approximation.

2. MAIN RESULTS

2.1. Notation and assumptions. All vector norms are Euclidean, and the norm on a product $(\mathbb{R}^d)^k$ is the unnormalized one, $|x|^2 = \sum_{i=1}^k |x_i|^2$; matrix norms are operator norms unless indicated otherwise, and $|\cdot|_{\text{HS}}$ is the Hilbert–Schmidt norm. For $x \in (\mathbb{R}^d)^N$ we write $x_{1:m} = (x_1, \dots, x_m)$, and for a law ρ on $(\mathbb{R}^d)^N$ we write ρ_m or $\rho^{(m)}$ for the law of $x_{1:m}$. A law on $(\mathbb{R}^d)^N$ is exchangeable if it is invariant under permutations of the N coordinates. We denote by $\mathcal{P}_2(\mathbb{R}^n)$ the probability measures with finite second moment, by

$$W_2^2(\rho, \eta) = \inf_{\pi \in \Pi(\rho, \eta)} \int |x - y|^2 \pi(dx, dy)$$

the quadratic Wasserstein distance, by $\rho(f) = \int f d\rho$ the integral of f , and by $H(\rho|\eta) = \int \log(d\rho/d\eta) d\rho$ the relative entropy, with $H(\rho|\eta) = +\infty$ if $\rho \not\ll \eta$. The space $C_b^1(\mathbb{R}^n)$ consists of bounded C^1 functions with bounded gradient, C_b^2 of bounded C^2 functions with bounded first and second derivatives, and C_b^∞ of smooth functions which are bounded together with all their derivatives; $C_c^1(I)$ denotes compactly supported C^1 functions on an open interval I , and $C^{1,2}$ refers to functions of (u, x) that are C^1 in u and C^2 in x . For $K: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ of class C^1 , $D_x K$ and $D_y K$ are the Jacobian matrices in the first and in the second variable.

Assumption A. There are constants $L_a, L_1, L_2, M \geq 0$ such that for all $x, x', y, y' \in \mathbb{R}^d$,

$$(2.1) \quad |a(x) - a(x')| \leq L_a |x - x'|,$$

$$(2.2) \quad |K(x, y)| \leq M,$$

$$(2.3) \quad |K(x, y) - K(x', y')| \leq L_1 |x - x'| + L_2 |y - y'|.$$

2.2. The main theorem.

Theorem 2.1. Suppose that Assumption A holds. Let $N \geq 1$, let $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$, let $P_{N,0} \in \mathcal{P}_2((\mathbb{R}^d)^N)$ be exchangeable, and let $C_0 \geq 0$ be such that

$$(2.4) \quad W_2^2(P_{N,0}^{(k)}, \mu_0^{\otimes k}) \leq C_0 \frac{k^2}{N^2}, \quad 1 \leq k \leq N.$$

Let $P_{N,t}$ and μ_t be the laws at time t of the solutions of (1.1) and (1.2). Then, for every $T > 0$,

$$(2.5) \quad \sup_{0 \leq t \leq T} W_2^2(P_{N,t}^{(k)}, \mu_t^{\otimes k}) \leq C_T \frac{k^2}{N^2}, \quad 1 \leq k \leq N,$$

where C_T is given by (2.9).

The constants are

$$(2.6) \quad L = L_a + L_1, \quad \Lambda = L_a + L_1 + L_2, \quad L_c = 2L_1 + L_2,$$

$$(2.7) \quad G = 2L_1^2 + L_2^2 + 2M^2, \quad \lambda = 2G, \quad D = 2\Lambda + 2L_1 + 1, \quad \omega = D + 3\lambda,$$

$$(2.8) \quad D_T = \frac{1 + LT + L^2 T^2 / 3}{4}, \quad A_0 = 2M + L_c e^{LT} \sqrt{C_0}, \quad A_1 = \sqrt{8} M \sqrt{C_0 D_T},$$

$$(2.9) \quad C_T = \left[e^{LT} \sqrt{C_0} + e^{\omega T / 2} (A_0 T + 2A_1 \sqrt{T}) \right]^2.$$

Thus C_T depends only on C_0, T, L_a, L_1, L_2 and M . It does not depend on N or on the dimension d , and it involves neither $a(0)$, nor moments of $P_{N,0}$ and μ_0 , nor derivatives of the

coefficients. The bound (2.5) at level k uses (2.4) at all levels between k and N , through Proposition 3.7 and Lemma 4.2.

2.3. Consequences.

Corollary 2.2. *Suppose that Assumption A holds and $P_{N,0} = \mu_0^{\otimes N}$ with $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$. Then, for every $T > 0$,*

$$\sup_{0 \leq t \leq T} W_2^2(P_{N,t}^{(k)}, \mu_t^{\otimes k}) \leq 4M^2 T^2 e^{\omega T} \frac{k^2}{N^2}, \quad 1 \leq k \leq N.$$

Proof. This is Theorem 2.1 with $C_0 = 0$, for which $A_0 = 2M$, $A_1 = 0$ and $C_T = 4M^2 T^2 e^{\omega T}$. $\square$

In this case the reference law R_s^N of Section 3 equals $\mu_s^{\otimes N}$, the external term of Lemma 4.6 vanishes, and no entropy estimate enters the proof.

Corollary 2.3. *Under the hypotheses of Theorem 2.1, let $X^N(t)$ be the solution of (1.1) and $0 \leq t \leq T$. For $1 \leq k \leq N$, every Lipschitz function $F: (\mathbb{R}^d)^k \rightarrow \mathbb{R}$ satisfies*

$$(2.10) \quad \left| \mathbb{E}F(X_1^N(t), \dots, X_k^N(t)) - \int F d\mu_t^{\otimes k} \right| \leq \sqrt{C_T} \text{Lip}(F) \frac{k}{N}.$$

If $N \geq 2$, every bounded Lipschitz function $f: \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies

$$(2.11) \quad \left| \text{Cov}(f(X_1^N(t)), f(X_2^N(t))) \right| \leq \frac{4\sqrt{2} \sqrt{C_T} \|f\|_\infty \text{Lip}(f)}{N} + \frac{C_T \text{Lip}(f)^2}{N^2}.$$

Proof. The first Wasserstein distance is bounded by W_2 , so (2.10) follows from Theorem 2.1 and the Kantorovich–Rubinstein duality. For (2.11), write $X_i = X_i^N(t)$, $m = \mu_t(f)$ and $g(x_1, x_2) = (f(x_1) - m)(f(x_2) - m)$. Since X_1 and X_2 have the same law,

$$\text{Cov}(f(X_1), f(X_2)) = \mathbb{E}g(X_1, X_2) - (\mathbb{E}f(X_1) - m)^2.$$

The function g satisfies $\int g d\mu_t^{\otimes 2} = 0$ and

$$|g(x) - g(y)| \leq 2\|f\|_\infty \text{Lip}(f)(|x_1 - y_1| + |x_2 - y_2|) \leq 2\sqrt{2} \|f\|_\infty \text{Lip}(f) |x - y|,$$

so (2.10) with $k = 2$ gives $|\mathbb{E}g(X_1, X_2)| \leq 4\sqrt{2} \sqrt{C_T} \|f\|_\infty \text{Lip}(f)/N$, while (2.10) with $k = 1$ gives $|\mathbb{E}f(X_1) - m| \leq \sqrt{C_T} \text{Lip}(f)/N$. $\square$

The boundedness assumption concerns only the fluctuating part $y \mapsto K(x, y)$ of the interaction: a linear confinement may be placed inside the pairwise drift, whereas interactions growing in the second variable, such as $y - x$, are excluded.

Proposition 2.4. *Let $b: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ satisfy*

$$(2.12) \quad |b(x, y) - b(x', y')| \leq \ell_x |x - x'| + \ell_y |y - y'|, \quad \sup_{x, y, z} |b(x, y) - b(x, z)| \leq M_b.$$

Then Theorem 2.1 applies to the particle drift $N^{-1} \sum_j b(x_i, x_j)$ with $L_a = \ell_x$, $L_1 = 2\ell_x$, $L_2 = \ell_y$ and $M = M_b$. Conversely, if $b(x, y) = a(x) + K(x, y)$ with a and K satisfying Assumption A, then (2.12) holds with $\ell_x = L_a + L_1$, $\ell_y = L_2$ and $M_b = 2M$.

Proof. Put $a(x) = b(x, 0)$ and $K(x, y) = b(x, y) - b(x, 0)$. Then $|K| \leq M_b$, a is ℓ_x -Lipschitz, and $|K(x, y) - K(x', y')| \leq |b(x, y) - b(x', y')| + |b(x, 0) - b(x', 0)| \leq 2\ell_x |x - x'| + \ell_y |y - y'|$. Since the diagonal term is included and the normalization is $1/N$, we have $N^{-1} \sum_j a(x_i) = a(x_i)$, so the two particle systems coincide. The converse follows from the triangle inequality and (2.1)–(2.3). $\square$

The mean field normalization $(N-1)^{-1} \sum_{j \neq i}$ without diagonal term, used in [39], only changes the constant.

Proposition 2.5. *Let $N \geq 2$, and let $\tilde{P}_{N,t}$ be the law at time t of the system obtained from (1.1) by replacing $N^{-1} \sum_{j=1}^N K(X_i^N, X_j^N)$ with $(N-1)^{-1} \sum_{j \neq i} K(X_i^N, X_j^N)$, started from the same initial law. Under the hypotheses of Theorem 2.1,*

$$\sup_{0 \leq t \leq T} W_2(\tilde{P}_{N,t}^{(k)}, \mu_t^{\otimes k}) \leq (\sqrt{C_T} + 2MTe^{\Lambda T}) \frac{k}{N}, \quad 1 \leq k \leq N.$$

Proof. The drift of the modified system is $b_N + e$, where b_N is the drift of (1.1) and $e_i(x) = (N(N-1))^{-1} \sum_{j \neq i} K(x_i, x_j) - N^{-1} K(x_i, x_i)$, so that $|e_i| \leq 2M/N$ and $|e| \leq 2M/\sqrt{N}$. Couple both systems through the same initial vector and Brownian motions, and let $\Delta = \tilde{X} - X$. Since $\text{Lip}(b_N) \leq \Lambda$ by Lemma 3.1, $|\Delta_t| \leq \int_0^t (\Lambda |\Delta_u| + 2M/\sqrt{N}) du$, hence $|\Delta_t| \leq 2MTe^{\Lambda T}/\sqrt{N}$ for $t \leq T$. Both equations are invariant under permutations of the coordinates, and the initial law and the noise are exchangeable, so the $|\Delta_i(t)|^2$ are identically distributed and $\mathbb{E}|\Delta_{1:k}(t)|^2 = \frac{k}{N} \mathbb{E}|\Delta(t)|^2 \leq 4M^2T^2e^{2\Lambda T}k/N^2 \leq (2MTe^{\Lambda T}k/N)^2$. The triangle inequality and Theorem 2.1 conclude. $\square$

Remark 2.6. Theorem 2.1 also applies to narrowly continuous probability solutions $(P_{N,t})$ and (μ_t) of the forward Kolmogorov equations associated with (1.1) and (1.2), provided their second moments are locally bounded in time. Indeed, the drifts have linear growth, so by the superposition principle [60, 7] such solutions are time marginals of martingale solutions of the corresponding linear equations, and these are unique in law because the coefficients are Lipschitz [35, Sections 5.2–5.4]. Applied first to the drift B_t built from (μ_t) , this shows that (μ_t) solves (1.2), and both curves coincide with the laws of the strong solutions of Lemma 3.1. A constant noise intensity σdW with $\sigma > 0$ reduces to $\sqrt{2}dW$ through the time change $t \mapsto 2t/\sigma^2$, which multiplies a and K by $2/\sigma^2$.

2.4. Microcanonical initial laws. We write $\mathcal{N}_n$ for the standard Gaussian measure on $\mathbb{R}^n$ and take $\mu_0 = \mathcal{N}_d$, so that $\mu_0^{\otimes k} = \mathcal{N}_{dk}$. The following initial laws satisfy (2.4) and are singular with respect to $\mu_0^{\otimes N}$.

Proposition 2.7. *Let $Z_1, \dots, Z_N$ be independent with law $\mathcal{N}_d$, let $\bar{Z} = N^{-1} \sum_j Z_j$, and let $P_{N,0}$ be the law of $(Z_i - \bar{Z})_{1 \leq i \leq N}$. Then $P_{N,0}$ is exchangeable and concentrated on the subspace $\{x : \sum_i x_i = 0\}$, so that $H(P_{N,0} | \mu_0^{\otimes N}) = +\infty$, while for $1 \leq k < N$*

$$(2.13) \quad H(P_{N,0}^{(k)} | \mu_0^{\otimes k}) = \frac{d}{2} \left(-\frac{k}{N} - \log \left(1 - \frac{k}{N} \right) \right).$$

Moreover (2.4) holds with $C_0 = d$. If $a = 0$ and $K = 0$, then for all $t \geq 0$ and $1 \leq k \leq N$, with $v_t = 1 + 2t$,

$$(2.14) \quad W_2^2(P_{N,t}^{(k)}, \mu_t^{\otimes k}) = \frac{d(k/N)^2}{(\sqrt{v_t} + \sqrt{v_t - k/N})^2} \geq \frac{d}{4v_t} \frac{k^2}{N^2}.$$

Proof. Exchangeability is clear. The vector $(Z_i - \bar{Z})_{i \leq k}$ is centred Gaussian with covariance $\Sigma_k = (I_k - N^{-1} \mathbf{1} \mathbf{1}^\top) \otimes I_d$, where $\mathbf{1} \in \mathbb{R}^k$ is the vector of ones. The matrix Σ_k has the eigenvalue $1 - k/N$ with multiplicity d , on the span of the vectors $\mathbf{1} \otimes e$, $e \in \mathbb{R}^d$, and the eigenvalue 1 with multiplicity $d(k-1)$. For $k = N$ the eigenvalue 0 shows that $P_{N,0}$ is carried by the subspace $\{\sum_i x_i = 0\}$, which is Lebesgue-null, so $P_{N,0}$ and $\mu_0^{\otimes N}$ are mutually singular. For $k < N$, (2.13) is the formula $H(\mathcal{N}(0, \Sigma) | \mathcal{N}_n) = \frac{1}{2}(\text{tr } \Sigma - n - \log \det \Sigma)$. If

$a = K = 0$, then $X_i^N(t) = Z_i - \bar{Z} + \sqrt{2}W_i(t)$, so that $P_{N,t}^{(k)}$ is centred Gaussian with covariance $(v_t I_k - N^{-1} \mathbf{1}\mathbf{1}^\top) \otimes I_d$, while $\mu_t = \mathcal{N}(0, v_t I_d)$. The two covariance matrices commute, and the formula for the quadratic Wasserstein distance between Gaussian measures with commuting covariances [20, 25] gives $W_2^2 = d(\sqrt{v_t} - \sqrt{v_t - k/N})^2$, which is (2.14). At $t = 0$ this equals $d(1 - \sqrt{1 - k/N})^2 \leq dk^2/N^2$, because $1 - \sqrt{1 - x} \leq x$ for $0 \leq x \leq 1$. $\square$

Proposition 2.8. *Let $P_{N,0} = \varpi_N$ be the uniform probability measure on the sphere $\{x \in (\mathbb{R}^d)^N : \sum_{i=1}^N |x_i|^2 = dN\}$. Then ϖ_N is exchangeable and singular with respect to $\mu_0^{\otimes N}$, and*

$$(2.15) \quad W_2^2(\varpi_N^{(k)}, \mu_0^{\otimes k}) \leq 32 \frac{k^2}{N^2}, \quad 1 \leq k \leq N.$$

Proof. Permutations of the particles are orthogonal maps of $(\mathbb{R}^d)^N$ that preserve the sphere, so ϖ_N is exchangeable, and it is singular with respect to $\mu_0^{\otimes N}$ because the sphere is Lebesgue-null. Put $n = dN$ and $j = dk$, and let Z have law $\mathcal{N}_n$. Then $X = \sqrt{n} Z/|Z|$ has law ϖ_N , and $\varpi_N^{(k)}$ is the law of the first j coordinates of X .

If $4k \geq N$, the coupling (X, Z) gives

$$W_2^2(\varpi_N, \mathcal{N}_n) \leq \mathbb{E}(|Z| - \sqrt{n})^2 \leq \frac{1}{n} \mathbb{E}(|Z|^2 - n)^2 = 2,$$

since $|\sqrt{r} - \sqrt{n}| \leq |r - n|/\sqrt{n}$ for $r \geq 0$. Projection onto the first j coordinates does not increase W_2 , so $W_2^2(\varpi_N^{(k)}, \mathcal{N}_j) \leq 2 \leq 32k^2/N^2$.

Suppose now that $4k < N$. Then $N \geq 5$, and $p := (n - j)/2 \geq 2$. The vector $Y = X_{1:j}$ is rotation invariant in $\mathbb{R}^j$, and $|Y|^2/n = |Z_{1:j}|^2/|Z|^2$ has the Beta($j/2, p$) distribution, since $|Z_{1:j}|^2$ and $|Z|^2 - |Z_{1:j}|^2$ are independent with χ_j^2 and χ_{n-j}^2 laws. Writing the density of Y in polar coordinates, we find that Y has the density

$$y \mapsto \frac{\Gamma(n/2)}{\Gamma(p)(\pi n)^{j/2}} \left(1 - \frac{|y|^2}{n}\right)^{p-1}, \quad |y|^2 < n.$$

Moreover $\mathbb{E}|Y|^2 = j$, because the n coordinates of X are exchangeable and $|X|^2 = n$, and $\mathbb{E} \log(1 - |Y|^2/n) = \psi(p) - \psi(n/2)$, where $\psi = \Gamma'/\Gamma$; the latter follows by differentiating $\log \int_0^1 u^{p-1}(1-u)^{j/2-1} du = \log \Gamma(p) + \log \Gamma(j/2) - \log \Gamma(n/2)$ in p . A direct computation gives $H(\varpi_N^{(k)} | \mathcal{N}_j) = \mathcal{I}(p, j/2)$, where

$$\mathcal{I}(p, \beta) = \log \Gamma(p + \beta) - \log \Gamma(p) - \beta \log(p + \beta) + \beta + (p - 1)(\psi(p) - \psi(p + \beta)), \quad \beta \geq 0.$$

We use the inequalities

$$(2.16) \quad \psi(x) < \log x - \frac{1}{2x}, \quad \psi'(x) > \frac{1}{x} + \frac{1}{2x^2}, \quad x > 0.$$

The second follows from $\psi'(x) = \sum_{l \geq 0} (x+l)^{-2}$ [48, Section 5.15] and the inequality $\sum_{l \geq 0} g(l) - \frac{1}{2}g(0) > \int_0^\infty g$ for the strictly convex function $g(u) = (x+u)^{-2}$, which follows by comparing each trapezoid with the integral over $[l, l+1]$. The first follows from the second, because $x \mapsto \log x - \frac{1}{2x} - \psi(x)$ then has a negative derivative and tends to 0 as $x \rightarrow \infty$ [48, Section 5.11].

Now $\mathcal{I}(p, 0) = 0$ and, for $\beta \geq 0$ and $x = p + \beta$,

$$\partial_\beta \mathcal{I}(p, \beta) = \psi(x) - \log x - \frac{\beta}{x} + 1 - (p-1)\psi'(x) < -\frac{1}{2x} - \frac{\beta}{x} + 1 - (p-1)\left(\frac{1}{x} + \frac{1}{2x^2}\right) = \frac{\beta+1}{2x^2},$$

where we used (2.16), $p \geq 1$ and $(p-1)/x = 1 - (\beta+1)/x$. Since $x \geq p$, integration over $\beta \in [0, j/2]$ gives

$$H(\varpi_N^{(k)} | \mathcal{N}_j) < \frac{j^2/4 + j}{4p^2} = \frac{j^2 + 4j}{4(n-j)^2} < \frac{4}{9} \frac{j^2 + 4j}{n^2} \leq \frac{20}{9} \frac{k^2}{N^2},$$

because $n-j > 3n/4$, $j^2/n^2 = k^2/N^2$ and $4j/n^2 = 4k/(dN^2) \leq 4k^2/N^2$. Talagrand's inequality $W_2^2(\nu, \mathcal{N}_j) \leq 2H(\nu | \mathcal{N}_j)$ [59] gives $W_2^2(\varpi_N^{(k)}, \mathcal{N}_j) \leq \frac{40}{9} k^2/N^2$. $\square$

In both examples $H(P_{N,0} | \mu_0^{\otimes N}) = +\infty$, so the hypothesis [39, (2.6)] fails at level $k = N$, while Theorem 2.1 applies for all a and K satisfying Assumption A. Proposition 2.7 also shows that the order k^2/N^2 in Theorem 2.1 cannot be improved.

3. TANGENT ENERGIES AND THE STRUCTURE OF THE PROOF

3.1. The dynamics. Write

$$(3.1) \quad b_N(x)_i = a(x_i) + \frac{1}{N} \sum_{j=1}^N K(x_i, x_j), \quad \mathcal{L}_N = \Delta + b_N \cdot \nabla, \quad \mathcal{L}_0^N(s) = \Delta + \sum_{i=1}^N B_s(x_i) \cdot \nabla_i.$$

The diagonal term $j = i$ is always retained. We write S_r^N for the semigroup of (1.1) acting on functions and $S_r^{N,*}$ for its action on measures. The transition family of the linear equation $dY = B_s(Y)ds + \sqrt{2}dW$ is denoted $Q_{u,s}$; it acts on measures from the right.

Lemma 3.1. *Under Assumption A, equations (1.1) and (1.2) have unique strong solutions for initial laws in $\mathcal{P}_2$, and $t \mapsto P_{N,t}$, $t \mapsto \mu_t$ are continuous in W_2 . The map $(t, x) \mapsto B_t(x)$ is continuous, $\mu_t = \mu_0 Q_{0,t}$, and*

$$(3.2) \quad \text{Lip}(b_N) \leq \Lambda, \quad \text{Lip}(B_t) \leq L$$

uniformly in N and t . The law $P_{N,t}$ is exchangeable. If moreover $a, K \in C_b^\infty$, the flow $F_r(x, W)$ of (1.1) is smooth in x and $J_r = D_x F_r$ satisfies $\|J_r\|_{\text{op}} \leq e^{\Lambda r}$.

Proof. For $\eta, \tilde{\eta} \in \mathcal{P}_2(\mathbb{R}^d)$, (2.1) and (2.3) give

$$\left| a(x) + \int K(x, y) \eta(dy) - a(x') - \int K(x', y) \tilde{\eta}(dy) \right| \leq L|x - x'| + L_2 W_2(\eta, \tilde{\eta}),$$

and the drift has linear growth. Well-posedness of (1.2) in $\mathcal{P}_2$, with a W_2 -continuous flow of time marginals, is then classical; see [9, Theorem 4.21], and [58, Chapter I] for bounded interactions. Continuity of B follows from $|B_t(x) - B_s(x)| \leq L_2 W_2(\mu_t, \mu_s)$. The process X solves the linear equation with drift B , whose law is unique, so $\mu_t = \mu_0 Q_{0,t}$.

For $u = x - x'$, the i -th component of $b_N(x) - b_N(x')$ is bounded by $(L_a + L_1)|u_i| + L_2 N^{-1} \sum_j |u_j|$; Minkowski's inequality in the product norm and $\sum_j |u_j| \leq \sqrt{N}|u|$ give $\text{Lip}(b_N) \leq \Lambda$. Integrating (2.3) in y gives $\text{Lip}(B_t) \leq L$. Hence (1.1) has a unique strong solution, and its law is exchangeable because the equation and the law of the driving noise are invariant under permutations of the coordinates. For smooth coefficients with bounded derivatives the flow is smooth in x [36, Chapter 4], and $J_r = I + \int_0^r D b_N(F_u) J_u du$ with $\|D b_N\|_{\text{op}} \leq \Lambda$, so Gronwall's inequality gives the bound on J_r . $\square$

3.2. Tangent energies. Let $\rho \in \mathcal{P}(\mathbb{R}^n)$ and let $\mathcal{H}_\rho$ be the closure in $L^2(\rho; \mathbb{R}^n)$ of $\{\nabla\varphi : \varphi \in C_c^\infty(\mathbb{R}^n)\}$, with orthogonal projection Π_ρ . For $V \in L^2(\rho; \mathbb{R}^n)$ we define the linear functional $\sigma = -\operatorname{div}(\rho V)$ on $C_b^1(\mathbb{R}^n)$ by

$$\sigma(\varphi) = \int \nabla\varphi \cdot V \, d\rho,$$

and call it a source carried by ρ and represented by V . Its tangent energy is

$$(3.3) \quad \mathcal{E}_\rho(\sigma) = \sup_{\varphi \in C_c^\infty(\mathbb{R}^n)} \left(2\sigma(\varphi) - \int |\nabla\varphi|^2 \, d\rho \right).$$

In the language of [1, Section 8.4], $\mathcal{H}_\rho$ is the tangent space of $\mathcal{P}_2(\mathbb{R}^n)$ at ρ , a source is a tangent vector written in divergence form, and $\mathcal{E}_\rho(\sigma)$ is the square of its norm, that is, of the $\dot{H}^{-1}$ norm weighted by ρ ; see also [49, 42, 52].

Lemma 3.2. *Let σ be a source carried by ρ and represented by V , and let $w = \Pi_\rho V$. Then $\nabla\varphi \in \mathcal{H}_\rho$ for every $\varphi \in C_b^1(\mathbb{R}^n)$, and $\sigma(\varphi) = \int \nabla\varphi \cdot w \, d\rho$ for all such φ . Moreover w is the only element of $\mathcal{H}_\rho$ with this property for all $\varphi \in C_c^\infty$,*

$$(3.4) \quad \mathcal{E}_\rho(\sigma) = \|w\|_{L^2(\rho)}^2, \quad \|V\|_{L^2(\rho)}^2 = \mathcal{E}_\rho(\sigma) + \|V - w\|_{L^2(\rho)}^2,$$

the supremum in (3.3) is unchanged if C_c^∞ is replaced by C_b^1 , and $|\sigma(\varphi)| \leq \mathcal{E}_\rho(\sigma)^{1/2} \|\nabla\varphi\|_{L^2(\rho)}$ for $\varphi \in C_b^1$.

Proof. Let $\chi \in C_c^\infty(\mathbb{R}^n)$ be equal to 1 on the unit ball and put $\chi_R = \chi(\cdot/R)$. For $\varphi \in C_b^1$, $\nabla(\chi_R\varphi) = \chi_R\nabla\varphi + \varphi\nabla\chi_R \rightarrow \nabla\varphi$ in $L^2(\rho)$ by dominated convergence, since $|\varphi\nabla\chi_R| \leq \|\varphi\|_\infty \|\nabla\chi\|_\infty/R$. Mollifications of $\chi_R\varphi$ have gradients converging uniformly, with supports in a fixed compact set, so $\nabla(\chi_R\varphi) \in \mathcal{H}_\rho$ and hence $\nabla\varphi \in \mathcal{H}_\rho$. Consequently $\sigma(\varphi) = \langle V, \nabla\varphi \rangle = \langle w, \nabla\varphi \rangle$, and an element of $\mathcal{H}_\rho$ is determined by its pairings with the dense set ∇C_c^∞ . Since $2\langle w, g \rangle - \|g\|^2 = \|w\|^2 - \|w - g\|^2$ for $g \in \mathcal{H}_\rho$, the supremum of this expression over the dense subset ∇C_c^∞ of $\mathcal{H}_\rho$, or over the intermediate set $\nabla C_b^1 \subset \mathcal{H}_\rho$, equals $\|w\|^2$. The second identity in (3.4) is Pythagoras' theorem, and the last bound is the Cauchy–Schwarz inequality. $\square$

Lemma 3.3. *Let $\rho \in \mathcal{P}((\mathbb{R}^d)^N)$, $V \in L^2(\rho; (\mathbb{R}^d)^N)$, and for $1 \leq m \leq N$ define*

$$(3.5) \quad \sigma_m(\varphi) = \int \sum_{i=1}^m \nabla_i \varphi(x_{1:m}) \cdot V_i(x) \, \rho(dx), \quad \varphi \in C_b^1((\mathbb{R}^d)^m).$$

Then σ_m is a source carried by ρ_m , represented by the conditional expectation $\mathbb{E}_\rho[V_{1:m} | x_{1:m}]$. Let $w_m = \Pi_{\rho_m} \mathbb{E}_\rho[V_{1:m} | x_{1:m}]$ and $E_m = \mathcal{E}_{\rho_m}(\sigma_m)$. Then $E_m \leq \|V\|_{L^2(\rho)}^2$ and, for $m < N$,

$$(3.6) \quad \langle w_{m+1}, (w_m, 0) \rangle_{L^2(\rho_{m+1})} = E_m, \quad \|w_{m+1} - (w_m, 0)\|_{L^2(\rho_{m+1})}^2 = E_{m+1} - E_m.$$

In particular $E_m \leq E_{m+1}$.

Proof. The representation follows by conditioning on $x_{1:m}$, and $E_m \leq \|V\|^2$ by Lemma 3.2 and Jensen's inequality. A function $\varphi \in C_b^1((\mathbb{R}^d)^m)$, regarded as a function of $x_{1:m+1}$, has gradient $(\nabla\varphi, 0)$, so $\sigma_{m+1}(\varphi) = \sigma_m(\varphi)$, that is, $\langle w_{m+1}, (\nabla\varphi, 0) \rangle_{L^2(\rho_{m+1})} = \langle w_m, \nabla\varphi \rangle_{L^2(\rho_m)}$. Choose $\varphi_k \in C_c^\infty$ with $\nabla\varphi_k \rightarrow w_m$ in $L^2(\rho_m)$; since ρ_m is the marginal of ρ_{m+1} , $(\nabla\varphi_k, 0) \rightarrow (w_m, 0)$ in $L^2(\rho_{m+1})$, which gives the first identity in (3.6). As $\|(w_m, 0)\|_{L^2(\rho_{m+1})}^2 = E_m$, expanding the square gives the second. $\square$

The next lemma converts tangent energies into transport lengths. It is a form of the Benamou–Brenier formula [4] and relies on the characterization of absolutely continuous curves in $(\mathcal{P}_2, W_2)$ by the continuity equation [1, Theorem 8.3.1].

Lemma 3.4. *Let $(\rho_s)_{s \in [0, t]}$ be a W_2 -continuous curve in $\mathcal{P}_2(\mathbb{R}^n)$. Assume that for every $\varphi \in C_c^\infty(\mathbb{R}^n)$ the map $s \mapsto \rho_s(\varphi)$ is continuously differentiable on $(0, t)$ and*

$$(3.7) \quad \left| \frac{d}{ds} \rho_s(\varphi) \right| \leq g(s) \|\nabla \varphi\|_{L^2(\rho_s)}, \quad 0 < s < t,$$

where $g \geq 0$ is continuous on $(0, t]$ and integrable on $(0, t)$. Then $W_2(\rho_0, \rho_t) \leq \int_0^t g(s) ds$.

Proof. Fix $0 < s_1 < s_2 < t$, write $\ell_s(\varphi) = \frac{d}{ds} \rho_s(\varphi)$, and let $\mathbb{L} = L^2([s_1, s_2] \times \mathbb{R}^n, ds \rho_s(dx); \mathbb{R}^n)$. On the space $\mathcal{T}$ of finite sums $\xi = \sum_k \chi_k(s) \nabla \varphi_k(x)$ with $\chi_k \in C([s_1, s_2])$ and $\varphi_k \in C_c^\infty$, define $\Psi(\xi) = \int_{s_1}^{s_2} \sum_k \chi_k(s) \ell_s(\varphi_k) ds$. For each s , $\sum_k \chi_k(s) \ell_s(\varphi_k) = \ell_s(\sum_k \chi_k(s) \varphi_k)$ is bounded by $g(s) \|\xi(s, \cdot)\|_{L^2(\rho_s)}$, so by the Cauchy–Schwarz inequality $|\Psi(\xi)| \leq \mathcal{G} \|\xi\|_{\mathbb{L}}$ with $\mathcal{G}^2 = \int_{s_1}^{s_2} g^2$. Hence Ψ is well defined on the closure of $\mathcal{T}$ in $\mathbb{L}$, and the Riesz representation theorem provides v in this closure with $\|v\|_{\mathbb{L}} \leq \mathcal{G}$ and $\Psi(\xi) = \iint v \cdot \xi d\rho_s ds$. Taking $\xi = \chi(s) \nabla \varphi$ with $\chi \in C_c^1((s_1, s_2))$ gives

$$- \int_{s_1}^{s_2} \chi'(s) \rho_s(\varphi) ds = \int_{s_1}^{s_2} \chi(s) \int v_s \cdot \nabla \varphi d\rho_s ds,$$

so $\partial_s \rho_s + \operatorname{div}(\rho_s v_s) = 0$ in the sense of distributions on $(s_1, s_2) \times \mathbb{R}^n$, linear combinations of products $\chi(s) \varphi(x)$ being dense in $C_c^\infty((s_1, s_2) \times \mathbb{R}^n)$. Since $\int_{s_1}^{s_2} \|v_s\|_{L^2(\rho_s)} ds \leq (s_2 - s_1)^{1/2} \mathcal{G} < \infty$, [1, Theorem 8.3.1] shows that (ρ_s) is absolutely continuous on $[s_1, s_2]$ with metric derivative at most $\|v_s\|_{L^2(\rho_s)}$, whence

$$W_2(\rho_{s_1}, \rho_{s_2}) \leq \int_{s_1}^{s_2} \|v_s\|_{L^2(\rho_s)} ds \leq (s_2 - s_1)^{1/2} \mathcal{G} \leq (s_2 - s_1) \max_{[s_1, s_2]} g.$$

Summing over a partition of $[\varepsilon, t - \varepsilon]$ and letting the mesh tend to zero gives $W_2(\rho_\varepsilon, \rho_{t-\varepsilon}) \leq \int_\varepsilon^{t-\varepsilon} g$, because g is continuous on $[\varepsilon, t - \varepsilon]$. Letting $\varepsilon \downarrow 0$ and using the continuity of the curve at its endpoints completes the proof. $\square$

3.3. The reference evolution, the source and the interpolation. In this subsection Assumption A and (2.4) hold. For $0 \leq s \leq T$ let

$$R_s^N = P_{N,0} Q_{0,s}^{\otimes N}$$

be the law at time s of N independent solutions of the linear equation $dY = B_s(Y)ds + \sqrt{2}dW$ started from $P_{N,0}$. It belongs to $\mathcal{P}_2$, it is exchangeable, and its marginals are $R_s^{(m)} = P_{N,0}^{(m)} Q_{0,s}^{\otimes m}$.

Lemma 3.5. *For $0 \leq s \leq T$ and $1 \leq m \leq N$, $W_2(R_s^{(m)}, \mu_s^{\otimes m}) \leq e^{Ls} \sqrt{C_0} m/N$.*

Proof. By Lemma 3.1, $\mu_s^{\otimes m} = \mu_0^{\otimes m} Q_{0,s}^{\otimes m}$. Run an optimal coupling of $P_{N,0}^{(m)}$ and $\mu_0^{\otimes m}$ through the reference dynamics, with the same Brownian motion in each pair of coupled coordinates. The difference of two coupled coordinates solves $\frac{d}{dr}(Y_r - Y'_r) = B_r(Y_r) - B_r(Y'_r)$, so $|Y_s - Y'_s| \leq e^{Ls} |Y_0 - Y'_0|$, and (2.4) gives the claim. $\square$

Let

$$\bar{K}_s(x) = \int K(x, y) \mu_s(dy), \quad c_s(x, y) = K(x, y) - \bar{K}_s(x), \quad \alpha_m = 1 - \frac{m}{N}.$$

Then $\int c_s(x, y) \mu_s(dy) = 0$, $|c_s| \leq 2M$, and c_s is $2L_1$ -Lipschitz in x and L_2 -Lipschitz in y . Define $U_s: (\mathbb{R}^d)^N \rightarrow (\mathbb{R}^d)^N$ by

$$(3.8) \quad (U_s)_i(x) = B_s(x_i) - b_N(x)_i = -\frac{1}{N} \sum_{j=1}^N c_s(x_i, x_j),$$

so that $\mathcal{L}_0^N(s) - \mathcal{L}_N = U_s \cdot \nabla$. The field U_s is permutation equivariant, in the sense that $U_s(x_\pi) = U_s(x)_\pi$ for every permutation π , where $(x_\pi)_i = x_{\pi(i)}$, and $|U_s| \leq 2M\sqrt{N}$. We call $\zeta_s^N = -\operatorname{div}(R_s^N U_s)$ the source at time s , and write $\zeta_s^{(m)}$ for its marginal sources (3.5), which are carried by $R_s^{(m)}$. Section 4 is devoted to the proof of the following estimate.

Proposition 3.6. *For $0 < s \leq T$ and $1 \leq m \leq N$,*

$$\mathcal{E}_{R_s^{(m)}}(\zeta_s^{(m)}) \leq A(s) \frac{m^2}{N^2}, \quad A(s) = \left(A_0 + \frac{A_1}{\sqrt{s}}\right)^2.$$

The source is transported by the linearization of the particle flow. Assume, in addition, that $a \in C_b^\infty(\mathbb{R}^d; \mathbb{R}^d)$ and $K \in C_b^\infty(\mathbb{R}^d \times \mathbb{R}^d; \mathbb{R}^d)$, and let $F_r(x, W)$ and $J_r = D_x F_r$ be as in Lemma 3.1. Let $\rho_{N,0} \in \mathcal{P}_2((\mathbb{R}^d)^N)$ be exchangeable and let $V \in L^2(\rho_{N,0}; (\mathbb{R}^d)^N)$ be permutation equivariant, that is, $V(x_\pi) = V(x)_\pi$ for $\rho_{N,0}$ -a.e. x and every permutation π . For $X_0 \sim \rho_{N,0}$ independent of W we set

$$(3.9) \quad \begin{aligned} \rho_{N,r} &= \operatorname{Law}(F_r(X_0, W)), \\ \sigma_{N,r}(\varphi) &= \mathbb{E}[\nabla \varphi(F_r(X_0, W)) \cdot J_r(X_0, W) V(X_0)], \quad \varphi \in C_b^1((\mathbb{R}^d)^N), \end{aligned}$$

so that $\sigma_{N,r}(\varphi)$ is the derivative at $\varepsilon = 0$ of $\mathbb{E}\varphi(F_r(X_0 + \varepsilon V(X_0), W))$. By Lemma 5.1, $\sigma_{N,r}$ is a source carried by $\rho_{N,r}$. Its marginal sources $\sigma_{m,r}$ are carried by the marginals $\rho_{m,r}$, and we put $E_m(r) = \mathcal{E}_{\rho_{m,r}}(\sigma_{m,r})$. The following proposition is proved in Section 5.

Proposition 3.7. *In this setting, if $E_m(0) \leq A m^2 / N^2$ for $1 \leq m \leq N$ and some $A \geq 0$, then*

$$E_m(r) \leq A e^{\omega r} \frac{m^2}{N^2}, \quad r \geq 0, \quad 1 \leq m \leq N.$$

Finally, for $0 < t \leq T$ we interpolate between $P_{N,t}$ and R_t^N by

$$(3.10) \quad \nu_s^{N,t} = S_{t-s}^{N,*} R_s^N, \quad 0 \leq s \leq t,$$

so that $\nu_0^{N,t} = P_{N,t}$ and $\nu_t^{N,t} = R_t^N$. For $0 < s < t$ we write $\rho_{m,r}^{[s]}$, $\sigma_{m,r}^{[s]}$ and $E_m^{[s]}(r)$ for the objects of (3.9) built from $\rho_{N,0} = R_s^N$ and $V = U_s$; these data are admissible, since R_s^N is exchangeable and U_s is bounded and permutation equivariant. Lemma 3.8 is proved in Section 6.

Lemma 3.8. *Assume that $a \in C_b^\infty$ and $K \in C_b^\infty$ satisfy Assumption A, and let $0 < t \leq T$. The curve $s \mapsto \nu_s^{N,t}$ is continuous in W_2 on $[0, t]$, and $\nu_s^{N,t} = \rho_{N,t-s}^{[s]}$ for $0 < s < t$. For $\varphi \in C_b^\infty((\mathbb{R}^d)^N)$ the map $s \mapsto \nu_s^{N,t}(\varphi)$ is continuously differentiable on $(0, t)$, and*

$$(3.11) \quad \frac{d}{ds} \nu_s^{N,t}(\varphi) = R_s^N(U_s \cdot \nabla S_{t-s}^N \varphi) = \sigma_{N,t-s}^{[s]}(\varphi).$$

3.4. Proof of Theorem 2.1. Suppose first that $a \in C_b^\infty$ and $K \in C_b^\infty$ satisfy Assumption A. Fix $0 < t \leq T$ and $1 \leq m \leq N$, and let $\nu_s^{N,t,(m)}$ be the m -th marginal of $\nu_s^{N,t}$. For $\varphi \in C_c^\infty((\mathbb{R}^d)^m)$, regarded as a function on $(\mathbb{R}^d)^N$, Lemma 3.8 shows that $s \mapsto \nu_s^{N,t,(m)}(\varphi)$ is continuously differentiable on $(0, t)$, with derivative $\sigma_{m,t-s}^{[s]}(\varphi)$, and $\sigma_{m,t-s}^{[s]}$ is a source carried by $\rho_{m,t-s}^{[s]} = \nu_s^{N,t,(m)}$. At $r = 0$ the transported source is $\sigma_{N,0}^{[s]} = \zeta_s^N$, so Proposition 3.6 gives $E_m^{[s]}(0) \leq A(s)m^2/N^2$ for all m , and Proposition 3.7 gives $E_m^{[s]}(t-s) \leq e^{\omega(t-s)} A(s)m^2/N^2$. By Lemma 3.2,

$$\left| \frac{d}{ds} \nu_s^{N,t,(m)}(\varphi) \right| \leq g(s) \|\nabla \varphi\|_{L^2(\nu_s^{N,t,(m)})}, \quad g(s) = e^{\omega(t-s)/2} \left(A_0 + \frac{A_1}{\sqrt{s}} \right) \frac{m}{N}.$$

The curve $s \mapsto \nu_s^{N,t,(m)}$ is continuous in W_2 by Lemma 3.8, and g is continuous on $(0, t]$ and integrable, so Lemma 3.4 gives

$$W_2(P_{N,t}^{(m)}, R_t^{(m)}) \leq \int_0^t g(s) ds \leq e^{\omega T/2} (A_0 T + 2A_1 \sqrt{T}) \frac{m}{N}.$$

Together with Lemma 3.5 at $s = t$, this proves (2.5) for $0 < t \leq T$. For $t = 0$, (2.5) is (2.4), since $C_T \geq C_0$.

For general coefficients, let a^n and K^n be the approximations of Lemma 7.1, and let $P_{N,t}^n$ and μ_t^n be the corresponding laws, with the same initial laws. The constants in Assumption A, and hence C_T , do not depend on n , so the smooth case gives $W_2(P_{N,t}^{n,(m)}, (\mu_t^n)^{\otimes m}) \leq \sqrt{C_T} m/N$. Projections do not increase W_2 , and the product coupling gives $W_2((\mu_t^n)^{\otimes m}, \mu_t^{\otimes m}) \leq \sqrt{m} W_2(\mu_t^n, \mu_t)$. Hence, by the triangle inequality and Lemma 7.2,

$$W_2(P_{N,t}^{(m)}, \mu_t^{\otimes m}) \leq \sqrt{C_T} \frac{m}{N} + W_2(P_{N,t}^n, P_{N,t}) + \sqrt{m} W_2(\mu_t^n, \mu_t) \longrightarrow \sqrt{C_T} \frac{m}{N}$$

as $n \rightarrow \infty$, uniformly in $t \leq T$. This proves Theorem 2.1.

4. ENTROPY AND THE SOURCE ESTIMATE

Throughout this section Assumption A and (2.4) hold; no smoothness of a and K is needed.

4.1. Entropy. The entropy–cost inequality below is proved by the coupling by change of measure of Arnaudon, Thalmaier and Wang [2]; see also [53, 61]. Its constant does not depend on the number of coordinates.

Lemma 4.1. *Let $\mathbf{b}: [0, s] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ be continuous, L -Lipschitz in space uniformly in time, and of linear growth, and let Q be the transition family of $dZ_r = \mathbf{b}_r(Z_r)dr + \sqrt{2}dW_r$. For $m \geq 1$ and $\rho, \eta \in \mathcal{P}_2((\mathbb{R}^d)^m)$,*

$$(4.1) \quad H(\rho Q_{0,s}^{\otimes m} | \eta Q_{0,s}^{\otimes m}) \leq \beta_L(s) W_2^2(\rho, \eta), \quad \beta_L(s) = \frac{1 + Ls + L^2 s^2/3}{4s}.$$

In particular $\beta_L(s) \leq D_T/s$ for $0 < s \leq T$.

Proof. The block drift on $(\mathbb{R}^d)^m$ is again L -Lipschitz, so it suffices to treat $m = 1$ in dimension dm . Fix x, y , put $u = y - x$ and $u_r = (1 - r/s)u$, and let $\tilde{\mathbf{b}}_r(z) = \mathbf{b}_r(z + u_r) + u/s$. If $\tilde{Z}$ solves the equation with drift $\tilde{\mathbf{b}}$ from x , then $\tilde{Z}_r + u_r$ solves the original equation from y , and $u_s = 0$; hence the law of $\tilde{Z}_s$ is $Q_{0,s}(y, \cdot)$. Moreover $|\mathbf{b}_r(z) - \tilde{\mathbf{b}}_r(z)| \leq (L(1 - r/s) + s^{-1})|u|$ for all r, z . Since this bound is deterministic, Novikov's condition holds, and Girsanov's theorem, together with uniqueness in law for both equations, gives for the path laws P_x and $\tilde{P}_{x,y}$ started at x

$$H(P_x | \tilde{P}_{x,y}) = \frac{1}{4} \mathbb{E}_{P_x} \int_0^s |\mathbf{b}_r(Z_r) - \tilde{\mathbf{b}}_r(Z_r)|^2 dr \leq \frac{1}{4} \int_0^s \left(L \left(1 - \frac{r}{s} \right) + \frac{1}{s} \right)^2 dr |u|^2 = \beta_L(s) |x - y|^2,$$

the factor $1/4$ reflecting the noise covariance $2I$. Projection to the endpoint does not increase entropy, so $H(Q_{0,s}(x, \cdot) | Q_{0,s}(y, \cdot)) \leq \beta_L(s) |x - y|^2$. Integrating against a coupling $\pi \in \Pi(\rho, \eta)$ and using the joint convexity of relative entropy gives (4.1) after taking the infimum over π . $\square$

The following two facts are standard; see [15] and [39, Section 4.3]. Let $\mu \in \mathcal{P}(\mathbb{R}^d)$ and let $\rho_{m+1}(dx, dy) = \rho_m(dx) \gamma_x(dy)$ be a law on $(\mathbb{R}^d)^m \times \mathbb{R}^d$. The chain rule reads

$$(4.2) \quad H(\rho_{m+1} | \mu^{\otimes(m+1)}) - H(\rho_m | \mu^{\otimes m}) = \int H(\gamma_x | \mu) \rho_m(dx)$$

whenever the left side is defined, and for every measurable $F: \mathbb{R}^d \rightarrow \mathbb{R}^d$ with $|F| \leq M$, Pinsker's inequality gives

$$(4.3) \quad \left| \int F d(\gamma - \mu) \right|^2 \leq 2M^2 H(\gamma | \mu).$$

Lemma 4.2. *Let $R \in \mathcal{P}((\mathbb{R}^d)^N)$ be exchangeable, $\mu \in \mathcal{P}(\mathbb{R}^d)$, $h_j = H(R_j | \mu^{\otimes j})$, $h_0 = 0$, and assume $h_N < \infty$. Then $\delta_j = h_j - h_{j-1}$ is nonnegative and nondecreasing in j . Consequently $\delta_{m+1} \leq h_{2m}/m$ if $2m \leq N$, and $\delta_{m+1} \leq h_N/(N-m)$ if $m < N$. If $h_j \leq H_* j^2/N^2$ for all j , then*

$$(4.4) \quad \left(1 - \frac{m}{N}\right)^2 m \delta_{m+1} \leq 4H_* \frac{m^2}{N^2}, \quad 1 \leq m < N.$$

Proof. All h_j are finite because entropy does not increase under projection, and $\delta_{m+1} = \mathbb{E} H(\text{Law}(X_{m+1} | X_{1:m}) | \mu) \geq 0$ by (4.2). The conditional law of X_{m+1} given $X_{1:m-1}$ is the conditional expectation, given $X_{1:m-1}$, of its conditional law given $X_{1:m}$, so by convexity of $H(\cdot | \mu)$,

$$\mathbb{E} H(\text{Law}(X_{m+1} | X_{1:m-1}) | \mu) \leq \delta_{m+1}.$$

By exchangeability the left side equals $\mathbb{E} H(\text{Law}(X_m | X_{1:m-1}) | \mu) = \delta_m$. Monotonicity gives $m\delta_{m+1} \leq \sum_{j=m+1}^{2m} \delta_j \leq h_{2m}$ and $(N-m)\delta_{m+1} \leq h_N$. For (4.4): if $2m \leq N$, then $m\delta_{m+1} \leq h_{2m} \leq 4H_* m^2/N^2$; if $2m > N$, then $(1 - m/N)^2 m\delta_{m+1} \leq H_* m(N-m)/N^2 \leq H_* m^2/N^2$. $\square$

For laws with densities, the monotonicity of δ_j is the relative entropy counterpart of the fact that $H(X_n | X_{n-1}, \dots, X_1)$ is nonincreasing in n for stationary sequences [15, Theorem 4.2.2].

4.2. Proof of Proposition 3.6. Fix $0 < s \leq T$, and recall the notation of Section 3.3.

Lemma 4.3. *For $1 \leq m \leq N$,*

$$(4.5) \quad h_m(s) := H(R_s^{(m)} | \mu_s^{\otimes m}) \leq C_0 \beta_L(s) \frac{m^2}{N^2} \leq \frac{C_0 D_T}{s} \frac{m^2}{N^2}.$$

Proof. Since $\mu_s^{\otimes m} = \mu_0^{\otimes m} Q_{0,s}^{\otimes m}$ and $R_s^{(m)} = P_{N,0}^{(m)} Q_{0,s}^{\otimes m}$, this is Lemma 4.1 with $\mathfrak{b} = B$, combined with (2.4). $\square$

In particular $h_N(s) < \infty$, even when $H(P_{N,0} | \mu_0^{\otimes N}) = +\infty$, as in Section 2.4.

Lemma 4.4. *For $\varphi \in C_b^2((\mathbb{R}^d)^N)$, $\zeta_s^N(\varphi) = R_s^N((\mathcal{L}_0^N(s) - \mathcal{L}_N)\varphi)$. For $1 \leq m \leq N$, the marginal source $\zeta_s^{(m)}$ is represented in $L^2(R_s^{(m)})$ by*

$$(4.6) \quad g_s^{(m)}(x)_i = -\frac{1}{N} \sum_{j=1}^m c_s(x_i, x_j) - \alpha_m \mathbb{E}_{R_s^N} [c_s(x_i, X_{m+1}) | X_{1:m} = x], \quad 1 \leq i \leq m,$$

where the second term is absent if $m = N$.

Proof. The Laplacians and the one-body drifts cancel in $\mathcal{L}_0^N(s) - \mathcal{L}_N$, which equals $U_s \cdot \nabla$; the representation (3.8) of U_s through c_s uses $\bar{K}_s(x_i) = N^{-1} \sum_j \bar{K}_s(x_j)$. By exchangeability of R_s^N , for each $j > m$ the conditional law of X_j given $X_{1:m}$ equals that of X_{m+1} , so $\mathbb{E}[(U_s)_i | X_{1:m} = x] = g_s^{(m)}(x)_i$, and Lemma 3.3 applies. $\square$

Lemma 4.5. *Let $\Xi_s(x)_i = \sum_{j=1}^m c_s(x_i, x_j)$ for $x \in (\mathbb{R}^d)^m$, and $I_m(s) = N^{-2} \int |\Xi_s|^2 dR_s^{(m)}$. Then $\sqrt{I_m(s)} \leq A_0 m/N$ for $1 \leq m \leq N$.*

Proof. Let $Y \sim \mu_s^{\otimes m}$. Conditionally on Y_i , the variables $c_s(Y_i, Y_j)$, $j \neq i$, are independent and centred, with conditional second moment at most $\mathbb{E}|K(Y_i, Y_j)|^2 \leq M^2$; the diagonal term satisfies $|c_s(Y_i, Y_i)| \leq 2M$, and its conditional covariance with the off-diagonal sum vanishes. Hence $\mathbb{E}|\Xi_s(Y)_i|^2 \leq 4M^2 + (m-1)M^2$ and $\mathbb{E}|\Xi_s(Y)|^2 \leq m(m+3)M^2 \leq 4M^2m^2$. For $x, y \in (\mathbb{R}^d)^m$ and $u = x - y$,

$$|\Xi_s(x)_i - \Xi_s(y)_i| \leq 2mL_1|u_i| + L_2 \sum_{j=1}^m |u_j|,$$

so $|\Xi_s(x) - \Xi_s(y)| \leq mL_c|u|$. For any coupling of $X \sim R_s^{(m)}$ and Y , Minkowski's inequality gives $\sqrt{I_m(s)} \leq N^{-1}(2Mm + mL_c \mathbb{E}[|X - Y|^2]^{1/2})$. Optimizing the coupling and using Lemma 3.5, $m \leq N$ and $s \leq T$ gives the claim. $\square$

Lemma 4.6. *For $m < N$ let $\mathcal{X}_m(s) = \alpha_m^2 \sum_{i=1}^m \mathbb{E}_{R_s^N} [\mathbb{E}_{R_s^N} [c_s(X_i, X_{m+1}) | X_{1:m}]]^2$, and let $\mathcal{X}_N(s) = 0$. Then*

$$\mathcal{X}_m(s) \leq 8M^2 C_0 \beta_L(s) \frac{m^2}{N^2} \leq \frac{A_1^2 m^2}{s N^2}, \quad 1 \leq m \leq N.$$

Proof. Under R_s^N , let γ_x denote the conditional law of X_{m+1} given $X_{1:m} = x$. Since $c_s(x_i, \cdot)$ has mean zero under μ_s ,

$$\mathbb{E}_{R_s^N} [c_s(x_i, X_{m+1}) | X_{1:m} = x] = \int K(x_i, y) (\gamma_x - \mu_s)(dy).$$

By (4.3), summation over $i \leq m$ and (4.2),

$$\mathcal{X}_m(s) \leq 2M^2 \alpha_m^2 m (h_{m+1}(s) - h_m(s)).$$

Lemma 4.2 with $H_* = C_0 \beta_L(s)$, which is admissible by (4.5), gives the first inequality, and $\beta_L(s) \leq D_T/s$ gives the second. $\square$

Proof of Proposition 3.6. By Lemma 3.2, the energy is at most $\|g_s^{(m)}\|_{L^2(R_s^{(m)})}^2$. The two terms in (4.6) have L^2 norms $\sqrt{I_m(s)}$ and $\sqrt{\mathcal{X}_m(s)}$, so Minkowski's inequality and Lemmas 4.5 and 4.6 give the claim. $\square$

5. THE TANGENT HIERARCHY

In this section $a \in C_b^\infty(\mathbb{R}^d; \mathbb{R}^d)$ and $K \in C_b^\infty(\mathbb{R}^d \times \mathbb{R}^d; \mathbb{R}^d)$ satisfy Assumption A, and $\rho_{N,0}$, V , $\rho_{m,r}$, $\sigma_{m,r}$ and $E_m(r)$ are as in Section 3.3. The constants D , λ and ω do not depend on $\|a\|_\infty$ or on derivatives of order higher than one.

5.1. Marginal equations. For $1 \leq m \leq N$, $x \in (\mathbb{R}^d)^m$ and $y \in \mathbb{R}^d$, put

$$(5.1) \quad b_i^{[m]}(x) = a(x_i) + \frac{1}{N} \sum_{j=1}^m K(x_i, x_j), \quad \mathcal{L}_m = \Delta + b^{[m]} \cdot \nabla,$$

$$\Phi\psi(x, y) = \sum_{i=1}^m K(x_i, y) \cdot \nabla_i \psi(x).$$

Lemma 5.1. *The source $\sigma_{N,r}$ is carried by $\rho_{N,r}$ and represented by $\mathbb{E}[J_r V(X_0) | F_r(X_0, W) = \cdot]$; it is exchangeable in the sense that $\sigma_{N,r}(\varphi \circ \pi) = \sigma_{N,r}(\varphi)$ for every permutation π of the coordinates. In particular*

$$(5.2) \quad E_m(r) \leq e^{2\Lambda r} \|V\|_{L^2(\rho_{N,0})}^2.$$

For $\psi \in C_b^\infty((\mathbb{R}^d)^m)$ the maps $r \mapsto \rho_{m,r}(\psi)$ and $r \mapsto \sigma_{m,r}(\psi)$ are continuously differentiable, and for $m < N$

$$(5.3) \quad \frac{d}{dr} \rho_{m,r}(\psi) = \rho_{m,r}(\mathcal{L}_m \psi) + \alpha_m \rho_{m+1,r}(\Phi \psi),$$

$$(5.4) \quad \frac{d}{dr} \sigma_{m,r}(\psi) = \sigma_{m,r}(\mathcal{L}_m \psi) + \alpha_m \sigma_{m+1,r}(\Phi \psi),$$

while for $m = N$ the terms with $\alpha_N = 0$ are absent.

Proof. Conditioning gives the representation, and its L^2 norm is at most $e^{\Lambda r} \|V\|$ by Lemma 3.1; (5.2) then follows from Lemma 3.3. Exchangeability follows from $F_r(x_\pi, W_\pi) = F_r(x, W)_\pi$, the corresponding identity for J_r , the exchangeability of X_0 and W , and the equivariance of V .

By Itô's formula, $\frac{d}{dr} \rho_{N,r}(\varphi) = \rho_{N,r}(\mathcal{L}_N \varphi)$ for $\varphi \in C_b^\infty$. With $Z_r = J_r V(X_0)$ we have $dZ_r = Db_N(F_r)Z_r dr$, and Itô's formula applied to $\nabla \varphi(F_r) \cdot Z_r$, with the noise covariance $2I$, gives

$$\frac{d}{dr} \sigma_{N,r}(\varphi) = \mathbb{E}[(D^2 \varphi b_N + \nabla \Delta \varphi + (Db_N)^\top \nabla \varphi)(F_r) \cdot Z_r] = \sigma_{N,r}(\mathcal{L}_N \varphi),$$

the stochastic integral being a martingale because all integrands are bounded by constants times $e^{\Lambda r} |V(X_0)|$. The right sides are continuous in r by dominated convergence. For ψ depending on $x_{1:m}$ we have $\mathcal{L}_N \psi = \mathcal{L}_m \psi + N^{-1} \sum_{j>m} \Phi \psi(x_{1:m}, x_j)$, and exchangeability of $\rho_{N,r}$ and $\sigma_{N,r}$ replaces each term $j > m$ by the term $j = m+1$. This gives (5.3) and (5.4). $\square$

5.2. Pointwise identities.

Lemma 5.2. *Let $f \in C_b^\infty((\mathbb{R}^d)^m)$ and set $\bar{A}_m = \sqrt{m}(\|a\|_\infty + M)$. Then, pointwise,*

$$(5.5) \quad 2\nabla f \cdot \nabla \Delta f - \Delta |\nabla f|^2 = -2|D^2 f|_{\text{HS}}^2,$$

$$(5.6) \quad 2\nabla f \cdot \nabla (b^{[m]} \cdot \nabla f) - b^{[m]} \cdot \nabla |\nabla f|^2 = 2\nabla f \cdot (Db^{[m]})^\top \nabla f \leq 2\Lambda |\nabla f|^2,$$

$$(5.7) \quad 2\nabla f \cdot \nabla_x \Phi f - \Phi(|\nabla f|^2) = 2 \sum_{i=1}^m \nabla_i f \cdot (D_x K(x_i, y))^\top \nabla_i f \leq 2L_1 |\nabla f|^2,$$

$$(5.8) \quad |\nabla_{(x,y)} \Phi f|^2 \leq mG(|\nabla f|^2 + |D^2 f|_{\text{HS}}^2),$$

$$(5.9) \quad |\nabla(b^{[m]} \cdot \nabla f)| \leq \Lambda |\nabla f| + \bar{A}_m |D^2 f|_{\text{HS}}.$$

Proof. Identity (5.5) is Bochner's formula in flat space. Since $\nabla(b^{[m]} \cdot \nabla f) = (Db^{[m]})^\top \nabla f + D^2 f b^{[m]}$ and $\nabla |\nabla f|^2 = 2D^2 f \nabla f$, the Hessian terms cancel in (5.6); the estimate $\|Db^{[m]}\|_{\text{op}} \leq \Lambda$ follows exactly as the bound on $\text{Lip}(b_N)$ in Lemma 3.1, and $|b^{[m]}| \leq \bar{A}_m$ gives (5.9). Next,

$$\nabla_{x_i} \Phi f = (D_x K(x_i, y))^\top \nabla_i f + \sum_{k=1}^m D_{ik}^2 f K(x_k, y), \quad \Phi(|\nabla f|^2) = 2 \sum_{i,k=1}^m \nabla_i f \cdot D_{ik}^2 f K(x_k, y),$$

where $D_{ik}^2 f$ is the block $\partial_{x_i} \partial_{x_k} f$; this gives (5.7), with $\|D_x K\|_{\text{op}} \leq L_1$. From the same formula, $|\nabla_x \Phi f|^2 \leq 2L_1^2 |\nabla f|^2 + 2|D^2 f|_{\text{HS}}^2 |(K(x_k, y))_k|^2 \leq 2L_1^2 |\nabla f|^2 + 2mM^2 |D^2 f|_{\text{HS}}^2$, while $\nabla_y \Phi f = \sum_k (D_y K(x_k, y))^\top \nabla_k f$ gives $|\nabla_y \Phi f|^2 \leq mL_2^2 |\nabla f|^2$. Since $m \geq 1$, adding yields (5.8). $\square$

5.3. The integrated hierarchy. Fix $1 \leq m \leq N$, and let $(\tau_k)_{k \geq 1}$ be the trigonometric functions on $(\mathbb{R}^d)^m$ introduced in Appendix A, with $\Delta \tau_k = -\kappa_k \tau_k$. For $j \geq 1$ and $r \geq 0$, define $G(r)$, $l(r)$, $c(r)$, f and $\epsilon_{m,j}(r)$ by (A.1) with $\rho = \rho_{m,r}$ and $\sigma = \sigma_{m,r}$, and let $w_m(r) \in \mathcal{H}_{\rho_{m,r}}$ be

the representative of $\sigma_{m,r}$. By the definition (3.5), $\sigma_{m+1,r}(\varphi) = \sigma_{m,r}(\varphi)$ for $\varphi \in C_b^1((\mathbb{R}^d)^m)$, so Lemma A.2, applied with $\rho' = \rho_{m+1,r}$ and $\sigma' = \sigma_{m+1,r}$, gives

$$(5.10) \quad \epsilon_{m,j} = \rho_{m,r}(|\nabla f|^2) + \eta_j |c|^2, \quad E_m - \epsilon_{m,j} = \|w_m - \nabla f\|_{L^2(\rho_{m,r})}^2 + \eta_j |c|^2,$$

$$(5.11) \quad \|w_{m+1} - (\nabla f, 0)\|_{L^2(\rho_{m+1,r})}^2 \leq E_{m+1} - \epsilon_{m,j} \quad \text{if } m < N,$$

together with $0 \leq \epsilon_{m,j}(r) \leq E_m(r)$ and $\epsilon_{m,j}(r) \rightarrow E_m(r)$ as $j \rightarrow \infty$.

Lemma 5.3. *The functions $\epsilon_{m,j}$ are continuously differentiable, and with $q_m = \lambda m \alpha_m$ and $C_m = 2(\Lambda^2 + \bar{A}_m^2)$,*

$$(5.12) \quad \epsilon'_{m,j} \leq D \epsilon_{m,j} + q_m (E_{m+1} - \epsilon_{m,j}) + C_m (E_m - \epsilon_{m,j}), \quad m < N,$$

and $\epsilon'_{N,j} \leq D \epsilon_{N,j} + C_N (E_N - \epsilon_{N,j})$.

Proof. By Lemma 5.1, G and I are continuously differentiable, hence so is $\epsilon_{m,j} = I^\top (G + \eta_j I)^{-1} I$, and

$$\epsilon'_{m,j} = 2I' \cdot c - c^\top G' c.$$

Freezing f at time r and inserting (5.3)–(5.4), with $I' \cdot c = \frac{d}{dr} \sigma_{m,r}(f)$ and $c^\top G' c = \frac{d}{dr} \rho_{m,r}(|\nabla f|^2)$, we obtain $\epsilon'_{m,j} = I + II + \alpha_m III$, where

$$\begin{aligned} I &= 2\sigma_{m,r}(\Delta f) - \rho_{m,r}(\Delta |\nabla f|^2), & II &= 2\sigma_{m,r}(b^{[m]} \cdot \nabla f) - \rho_{m,r}(b^{[m]} \cdot \nabla |\nabla f|^2), \\ III &= 2\sigma_{m+1,r}(\Phi f) - \rho_{m+1,r}(\Phi(|\nabla f|^2)). \end{aligned}$$

Write $\epsilon = \epsilon_{m,j}$, $\vartheta = E_m - \epsilon$, $P = \rho_{m,r}(|\nabla f|^2) \leq \epsilon$ and $H = \rho_{m,r}(|D^2 f|_{\text{HS}}^2)$.

Since $\Delta f = -\sum_k \kappa_k c_k \tau_k$ lies in the span of $\tau_1, \dots, \tau_j$, the normal equation $(G + \eta_j I)c = I$ gives $\sigma_{m,r}(\Delta f) = \rho_{m,r}(\nabla f \cdot \nabla \Delta f) - \eta_j \sum_k \kappa_k c_k^2$, so by (5.5), $I = -2H - 2\eta_j \sum_k \kappa_k c_k^2 \leq -2H$.

For II write $w_m = \nabla f + v$, so that $\|v\|_{L^2(\rho_{m,r})}^2 \leq \vartheta$ by (5.10). By (5.6) and (5.9),

$$II \leq 2\Lambda P + 2\sqrt{\vartheta}(\Lambda\sqrt{P} + \bar{A}_m\sqrt{H}) \leq \left(2\Lambda + \frac{1}{2}\right)P + \frac{1}{2}H + C_m\vartheta.$$

For III write $w_{m+1} = (\nabla f, 0) + z$, so that $\|z\|_{L^2(\rho_{m+1,r})}^2 \leq E_{m+1} - \epsilon$ by (5.11). By (5.7), (5.8) and Young's inequality,

$$III \leq 2L_1 P + 2\sqrt{E_{m+1} - \epsilon} \sqrt{mG(P + H)} \leq \left(2L_1 + \frac{1}{2}\right)P + \frac{1}{2}H + 2mG(E_{m+1} - \epsilon).$$

Multiplying by $\alpha_m \leq 1$ and adding, we find that the coefficient of H is at most -1 and the coefficient of $P \leq \epsilon$ is at most D , while $2mG\alpha_m = q_m$. This proves (5.12). For $m = N$ the term III is absent. $\square$

Proposition 5.4. *The energies E_m are measurable and locally bounded. For $0 \leq r_0 \leq r$ and $m < N$,*

$$(5.13) \quad E_m(r) \leq e^{(D-q_m)(r-r_0)} E_m(r_0) + q_m \int_{r_0}^r e^{(D-q_m)(r-v)} E_{m+1}(v) dv,$$

and $E_N(r) \leq e^{D(r-r_0)} E_N(r_0)$.

Proof. Measurability holds because E_m is a pointwise limit of the continuous functions $\epsilon_{m,j}$, and local boundedness follows from (5.2). Integrating (5.12) against the factor $e^{(D-q_m)(r-v)}$ gives

$$\epsilon_{m,j}(r) \leq e^{(D-q_m)(r-r_0)} \epsilon_{m,j}(r_0) + \int_{r_0}^r e^{(D-q_m)(r-v)} (q_m E_{m+1}(v) + C_m (E_m(v) - \epsilon_{m,j}(v))) dv.$$

As $j \rightarrow \infty$, $\epsilon_{m,j} \rightarrow E_m$ pointwise and $0 \leq E_m - \epsilon_{m,j} \leq E_m$ is bounded on $[r_0, r]$, so dominated convergence gives (5.13). The level N is treated in the same way with $q_N = 0$. $\square$

The constant C_m , which depends on $\|a\|_\infty$, does not survive the limit $j \rightarrow \infty$.

5.4. Proof of Proposition 3.7. Let $u_m(r) = Ae^{\omega r} m^2 / N^2$. Since $q_m \leq \lambda m$ and $\omega - D = 3\lambda$,

$$u'_m - (D - q_m)u_m - q_m u_{m+1} = \frac{Ae^{\omega r}}{N^2} (3\lambda m^2 - q_m(2m+1)) \geq 0,$$

so by variation of constants $u_m(r) \geq e^{(D-q_m)r} u_m(0) + q_m \int_0^r e^{(D-q_m)(r-v)} u_{m+1}(v) dv$ for $m < N$, and $u_N(r) \geq e^{Dr} u_N(0)$. We prove $E_m \leq u_m$ by descending induction on m . At $m = N$ this follows from Proposition 5.4 with $r_0 = 0$. If $E_{m+1} \leq u_{m+1}$, then (5.13) with $r_0 = 0$, the positivity of its kernel, $E_m(0) \leq u_m(0)$ and the preceding inequality give $E_m \leq u_m$. $\square$

Remark 5.5. Let Z be the pure-birth chain on $\{1, \dots, N\}$ that jumps from j to $j+1$ at rate q_j , and put $\tilde{u}_m(r) = Ae^{Dr} \mathbb{E}_m[Z_r^2] / N^2$. By the backward equation of Z , $\tilde{u}$ satisfies (5.13) with $r_0 = 0$ and with equality, and the induction above, with $\tilde{u}$ in place of u , gives $E_m \leq \tilde{u}_m$. Since $q_j \leq \lambda j \leq \lambda k$ for $j \leq k$, the chain Z can be coupled with a Yule process Y of rates λk in such a way that Z jumps only when Y does; then $Z_r \leq \min(Y_r, N)$, and $\mathbb{E}_m Y_r^2 = m^2 e^{2\lambda r} + m(e^{2\lambda r} - e^{\lambda r}) \leq m^2 e^{3\lambda r}$. Hence

$$E_m(r) \leq Ae^{Dr} \min(m^2 e^{3\lambda r}, N^2) / N^2.$$

The same comparison for the entropy hierarchy is discussed in [41, Section 1.5]. The rates q_m are of order m . Because the coupling term in (5.12) is $E_{m+1} - \epsilon_{m,j}$ rather than E_{m+1} , they act as jump rates of Z and not as growth rates of E_m .

6. THE INTERPOLATING CURVE

Proof of Lemma 3.8. For smooth coefficients with bounded derivatives, the flow $F_u(x, W)$ is twice continuously differentiable in x , with first and second derivatives bounded in every L^p uniformly in $u \leq T$ [36, Chapter 4]. Hence, for $\varphi \in C_b^\infty$, $S_u^N \varphi \in C_b^2$ with bounds uniform in $u \leq T$, its derivatives are continuous in (u, x) , and $\nabla S_u^N \varphi(x) = \mathbb{E}[J_u(x, W)^\top \nabla \varphi(F_u(x, W))]$. By the Markov property and Itô's formula, $S_{u+h}^N \varphi - S_u^N \varphi = \int_u^{u+h} S_v^N(\mathcal{L}_N \varphi) dv = \int_0^h S_v^N(\mathcal{L}_N S_u^N \varphi) dv$, so $u \mapsto S_u^N \varphi(x)$ is continuously differentiable with $\partial_u S_u^N \varphi = S_u^N(\mathcal{L}_N \varphi) = \mathcal{L}_N S_u^N \varphi$; thus $(u, x) \mapsto S_u^N \varphi(x)$ is of class $C^{1,2}$ with bounded derivatives on $[0, T] \times (\mathbb{R}^d)^N$.

Let Y be the reference process with $\text{Law}(Y_0) = P_{N,0}$, so that $\nu_s^{N,t}(\varphi) = \mathbb{E}[\psi_s(Y_s)]$ with $\psi_s = S_{t-s}^N \varphi$. Itô's formula gives

$$\frac{d}{ds} \mathbb{E}[\psi_s(Y_s)] = \mathbb{E}[(\partial_s \psi_s + \mathcal{L}_0^N(s) \psi_s)(Y_s)] = \mathbb{E}[(\mathcal{L}_0^N(s) - \mathcal{L}_N) \psi_s](Y_s) = R_s^N(U_s \cdot \nabla \psi_s),$$

and the formula for $\nabla \psi_s$ yields

$$R_s^N(U_s \cdot \nabla \psi_s) = \mathbb{E}[\nabla \varphi(F_{t-s}(Y_s, W)) \cdot J_{t-s}(Y_s, W) U_s(Y_s)] = \sigma_{N,t-s}^{[s]}(\varphi),$$

with W independent of Y_s . This is (3.11), and the same representation shows that $\nu_s^{N,t}$ is the law of $F_{t-s}(Y_s, W)$, that is, $\nu_s^{N,t} = \rho_{N,t-s}^{[s]}$. Continuity of the derivative in s follows by dominated convergence from the boundedness of U_s and $\nabla \varphi$, the bound on J , and the continuity of $s \mapsto Y_s$, $s \mapsto U_s$ and of the flow.

For the continuity of the curve, take $X_0 \sim P_{N,0}$ and independent Brownian motions W^1, W^2 on one probability space, let Θ_s be the flow of the reference dynamics from time 0 to s , and put

$$X_s = F_{t-s}(\Theta_s(X_0, W^1), W^2),$$

whose law is $\nu_s^{N,t}$. Both drifts are bounded by $\bar{M} = \|a\|_\infty + M$ in each coordinate, so for either flow, with fixed starting point and fixed noise,

$$\mathbb{E}|F_u - F_v|^2 \leq 2N\bar{M}^2|u - v|^2 + 4dN|u - v|.$$

Splitting $X_s - X_{s'}$ into a change of the starting point and a change of the interacting duration, and using (3.2), we obtain

$$\mathbb{E}|X_s - X_{s'}|^2 \leq 4N(e^{2\Lambda T} + 1)(\bar{M}^2|s - s'|^2 + 2d|s - s'|),$$

which proves that $s \mapsto \nu_s^{N,t}$ is continuous in W_2 . $\square$

7. SMOOTH APPROXIMATION

Lemma 7.1. *Under Assumption A there are $a^n \in C_b^\infty(\mathbb{R}^d; \mathbb{R}^d)$ and $K^n \in C_b^\infty(\mathbb{R}^d \times \mathbb{R}^d; \mathbb{R}^d)$ satisfying Assumption A with the same constants L_a, L_1, L_2, M , such that $a^n \rightarrow a$ locally uniformly, $K^n \rightarrow K$ uniformly, and $|a^n(x)| \leq |a(0)| + L_a(|x| + 1)$ for all n and x .*

Proof. Let η_ε be a smooth probability density supported in the ball of radius ε (in the relevant dimension), let $\theta^n(x)_l = n \sin(x_l/n)$ coordinatewise, and set $a^n = (a \circ \theta^n) * \eta_{1/n}$ and $K^n = K * \eta_{1/n}$. The map θ^n is 1-Lipschitz with bounded image and $|\theta^n(x)| \leq |x|$, so $a \circ \theta^n$ is bounded and L_a -Lipschitz; convolution preserves these properties as well as (2.2)–(2.3), and produces bounded derivatives of all orders. Since $\theta^n \rightarrow \text{id}$ locally uniformly, $a^n \rightarrow a$ locally uniformly, while $|K^n - K| \leq (L_1 + L_2)/n$. Finally $|\theta^n(x - z)| \leq |x| + 1$ for $|z| \leq 1/n$. $\square$

Lemma 7.2. *Let $P_{N,t}^n$ and μ_t^n be the laws of (1.1) and (1.2) with coefficients a^n, K^n from Lemma 7.1 and the same initial laws. For every fixed N ,*

$$\sup_{t \leq T} W_2(P_{N,t}^n, P_{N,t}) \rightarrow 0, \quad \sup_{t \leq T} W_2(\mu_t^n, \mu_t) \rightarrow 0.$$

Proof. Couple the particle systems through the same initial vector and Brownian motions, and let b_N^n be the particle drift built from a^n, K^n . Since $\text{Lip}(b_N^n) \leq \Lambda$, Gronwall's and the Cauchy–Schwarz inequalities give

$$\mathbb{E} \sup_{t \leq T} |X_t^n - X_t|^2 \leq T e^{2\Lambda T} \int_0^T \mathbb{E} |b_N^n(X_u) - b_N(X_u)|^2 du.$$

For fixed N the integrand tends to zero pointwise and is bounded by a constant times $N + |X_u|^2$, which is integrable uniformly in $u \leq T$; dominated convergence proves the first claim.

For the nonlinear equations, couple X^n and X through the same initial point and Brownian motion, let $(\bar{Y}^n, \bar{Y})$ be an independent copy of the pair, and put $z_n(t) = \mathbb{E}[\sup_{u \leq t} |X_u^n - X_u|^2]^{1/2}$ and

$$\varepsilon_n(u) = \|a^n(X_u) - a(X_u)\|_{L^2} + \|K^n(X_u, \bar{Y}_u) - K(X_u, \bar{Y}_u)\|_{L^2}.$$

Writing $B_u^n(X_u^n) - B_u(X_u) = \mathbb{E}'[K^n(X_u^n, \bar{Y}_u^n) - K^n(X_u, \bar{Y}_u)] + a^n(X_u^n) - a^n(X_u) + (a^n(X_u) - a(X_u) + \mathbb{E}'[K^n(X_u, \bar{Y}_u) - K(X_u, \bar{Y}_u)])$, where $\mathbb{E}'$ integrates out the copy, and using $\|\bar{Y}_u^n - \bar{Y}_u\|_{L^2} = \|X_u^n - X_u\|_{L^2}$, Minkowski's inequality gives $z_n(t) \leq \Lambda \int_0^t z_n + \int_0^t \varepsilon_n$. The growth bound of Lemma 7.1, local uniform convergence and the second moments of X give $\int_0^T \varepsilon_n \rightarrow 0$ by dominated convergence, and Gronwall's inequality concludes. $\square$

8. FURTHER DIRECTIONS

Interactions growing in the second variable. The constant M of (2.2) enters C_T through Lemmas 4.5 and 4.6 and through (5.8). If K is Lipschitz but unbounded, the first two can be adapted. In Lemma 4.5 one may use $|c_s(x, y)| \leq L_2 \int |y - y'| \mu_s(dy')$ and second moments of μ_s . In Lemma 4.6, since $y \mapsto e \cdot K(x, y)$ is L_2 -Lipschitz for every unit vector e , Pinsker's inequality can be replaced by a transport inequality $W_1(\nu, \mu_s)^2 \leq C H(\nu | \mu_s)$, which holds for some C if and only if μ_s has a Gaussian moment [6, 18]. In (5.8), however, $|K(x_k, y)|$ multiplies the Hessian of the test function and is integrated in y against the conditional law of a hidden particle given the observed ones; for unbounded K this calls for bounds on conditional second moments, which are not available for general carrying laws. Interactions that are linear in the second variable, as in the Gaussian example of [39, Example 2.8], are a natural first case.

Path space. The question in [39] concerns laws of trajectories on $[0, T]$. The interpolation (3.10) is built from the semigroups S^N and Q and acts on time marginals; a path-space version would require a tangent hierarchy for laws on $C([0, T]; \mathbb{R}^d)^k$.

Long times. By Remark 5.5, $E_m(r) \leq Ae^{Dr} \min(m^2 e^{3\lambda r}, N^2)/N^2$, since the comparison chain is absorbed at N . For a confining one-body drift one would like to replace D by a negative rate. The approximation of Section 7 replaces a by bounded drifts and destroys confinement, so this requires a version of Lemma 5.3 for unbounded drifts, in which C_m is not available.

Entropy at positive times. Theorem 2.1 gives no information in total variation. A bound $H(P_{N,t}^{(k)} | \mu_t^{\otimes k}) \leq C_t k^2 / N^2$ for $t > 0$ under (2.4) would give $\|P_{N,t}^{(k)} - \mu_t^{\otimes k}\|_{\text{TV}} = O(k/N)$ by Pinsker's inequality. The bound $A(s)$ of Proposition 3.6 is of order s^{-1} as $s \rightarrow 0$, and Section 3.4 integrates only its square root.

Non-exchangeable systems. The proof of Lemma 3.3 applies to any increasing sequence of coordinate sets. Exchangeability is used to reduce the $N - m$ hidden particles to a single one in Lemmas 4.4 and 5.1, and in Lemma 4.2. For heterogeneous interactions it would be natural to combine tangent energies with the subset-indexed hierarchy of [41].

APPENDIX A. GALERKIN APPROXIMATION

Fix $n \geq 1$ and enumerate the nonconstant real trigonometric functions $\cos(\xi \cdot x)$, $\sin(\xi \cdot x)$, $\xi \in \mathbb{Q}^n \setminus \{0\}$, as $(\tau_k)_{k \geq 1}$. Each τ_k belongs to $C_b^\infty(\mathbb{R}^n)$, and $\Delta \tau_k = -\kappa_k \tau_k$ with $\kappa_k = |\xi|^2$. In Section 5 these functions are used with $n = dm$.

Lemma A.1. *For every $\rho \in \mathcal{P}(\mathbb{R}^n)$, the gradients of finite linear combinations of the τ_k are dense in $\mathcal{H}_\rho$.*

Proof. These gradients lie in $\mathcal{H}_\rho$ by Lemma 3.2. Let $\varphi \in C_c^\infty$ and, for an integer p so large that $\text{supp } \varphi$ lies in the interior of $[-\pi p, \pi p]^n$, let φ_p be the $2\pi p$ -periodic extension of $\varphi|_{[-\pi p, \pi p]^n}$. Then $\|\nabla \varphi_p\|_\infty = \|\nabla \varphi\|_\infty$ and $\nabla \varphi_p \rightarrow \nabla \varphi$ pointwise, hence in $L^2(\rho)$. Fejér means of φ_p converge to φ_p in C^1 , and they are trigonometric polynomials with frequencies in $p^{-1}\mathbb{Z}^n \subset \mathbb{Q}^n$. $\square$

Let $\rho \in \mathcal{P}(\mathbb{R}^n)$, and let σ be a source carried by ρ with representative $w \in \mathcal{H}_\rho$. For $j \geq 1$ put $\eta_j = 1/(j+1)$ and

$$(A.1) \quad G_{pq} = \rho(\nabla \tau_p \cdot \nabla \tau_q), \quad l_p = \sigma(\tau_p), \quad c = (G + \eta_j I)^{-1} l, \quad f = \sum_{p=1}^j c_p \tau_p, \quad \epsilon_j = l \cdot c,$$

where $1 \leq p, q \leq j$.

Lemma A.2. *With this notation, $\epsilon_j = \rho(|\nabla f|^2) + \eta_j |c|^2$ and $\mathcal{E}_\rho(\sigma) - \epsilon_j = \|w - \nabla f\|_{L^2(\rho)}^2 + \eta_j |c|^2$; moreover $0 \leq \epsilon_j \leq \mathcal{E}_\rho(\sigma)$ and $\epsilon_j \rightarrow \mathcal{E}_\rho(\sigma)$ as $j \rightarrow \infty$. Let $\rho' \in \mathcal{P}(\mathbb{R}^n \times \mathbb{R}^{n'})$ have first marginal ρ , and let σ' be a source carried by ρ' , with representative $w' \in \mathcal{H}_{\rho'}$, such that $\sigma'(\varphi) = \sigma(\varphi)$ for every $\varphi \in C_b^1(\mathbb{R}^n)$, regarded as a function on $\mathbb{R}^n \times \mathbb{R}^{n'}$. Then $\|w' - (\nabla f, 0)\|_{L^2(\rho')}^2 \leq \mathcal{E}_{\rho'}(\sigma') - \epsilon_j$.*

Proof. The normal equation $(G + \eta_j I)c = \mathbf{l}$ gives $\sigma(f) = \mathbf{l} \cdot c = \epsilon_j = c^\top Gc + \eta_j |c|^2$, and $c^\top Gc = \rho(|\nabla f|^2)$; this is the first identity. Expanding $\|w - \nabla f\|^2 = \mathcal{E}_\rho(\sigma) - 2\sigma(f) + \rho(|\nabla f|^2)$ gives the second. Since $f \in C_b^1$, Lemma 3.2 gives $\langle w', (\nabla f, 0) \rangle_{L^2(\rho')} = \sigma'(f) = \sigma(f) = \epsilon_j$, and $\|(\nabla f, 0)\|_{L^2(\rho')}^2 = \rho(|\nabla f|^2)$ because ρ is the first marginal of ρ' ; expanding the square gives $\|w' - (\nabla f, 0)\|^2 = \mathcal{E}_{\rho'}(\sigma') - \epsilon_j - \eta_j |c|^2$. Finally, c maximizes $z \mapsto 2\mathbf{l} \cdot z - z^\top Gz - \eta_j |z|^2$ on $\mathbb{R}^j$, with maximal value $\epsilon_j \geq 0$. If $g = \sum_p z_p \tau_p$ is a fixed trigonometric polynomial, then for all large j the same completion of squares gives $0 \leq \mathcal{E}_\rho(\sigma) - \epsilon_j \leq \|w - \nabla g\|^2 + \eta_j |z|^2$. Letting $j \rightarrow \infty$ and then using Lemma A.1 proves the convergence. $\square$

STATEMENT ON THE USE OF AI

The author used Claude (Anthropic), a large language model, during the preparation of this manuscript, for editing the exposition and for checking computations, constants and references. The author verified all mathematical content and takes full responsibility for the paper.

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